

Syed Mohd Haider Rizvi

[in LinkedIn](#) | [+91-933-523-1073](#) | [smhrizvi281@gmail.com](#) | [GitHub](#)

Professional Summary

Passionate problem-solver since childhood - while others played with toys, I disassembled electronics to understand their inner workings, leading to repairing household devices and eventually pursuing software engineering to create maximum impact. **Senior Software Engineer with 4+ years** designing scalable systems at **25PB+ scale**. Proven track record: **5x API performance improvements**, **\$2.4M projected cost savings**, and **500K+ users impacted**. Creator of open-source tools serving the global developer community. Currently pursuing M.Tech in Applied AI while delivering enterprise-grade solutions.

Core Philosophy: "Technology should solve real problems and create meaningful impact at scale"

Technical Skills

- **Programming Languages:** Java, Python, SQL, C (foundational)
- **Cloud Platforms:** GCP, AWS
- **AI/ML:** LLMs, TensorFlow, GPT, LangChain
- **Frameworks:** Spring Boot, FastAPI, Docker, Kubernetes
- **Databases:** PostgreSQL, NoSQL, BigQuery
- **Expertise:** System Design, Microservices, Data Structures & Algorithms, Distributed Systems, DevOps (CI/CD)

Experience

Senior Software Engineer **FORD** *Bengaluru, KA, IN* **08/2024 - Present**

- **Architected 25PB+ scale data pipeline platforms** with cross-cloud portability (GCP/AWS), implementing microservices and distributed system patterns for enterprise reliability.
- **Built Agentic AI optimizer** leveraging LLMs + BigQuery, **projected \$2.4M annual savings** through automated resource allocation and query optimization algorithms.
- **Automated feed setup system** (Java + Spring Boot + K8s), reducing setup time from **5 hours → 30 seconds (99% reduction)**, saving **\$120K annually**.
- **Led global cross-functional teams**, reporting to Global Manager (USA), delivering scalable solutions across multiple time zones.

Senior Software Engineer **HSBC** *Pune, MH, IN* **01/2022 - 08/2024**

- **Developed NLP-powered AI solutions** using OpenAI's GPT models, integrated with enterprise data pipelines, enabling users to create SQL queries via natural language, reducing **development time by 80%** and improving data accessibility for financial reporting.
- **Optimized transaction APIs achieving 5x throughput improvement** through algorithmic optimization, async processing, database query optimization and caching strategies, impacting **300K+ daily users**.
- Redesigned **microservice architecture**, reducing **infrastructure costs by 20%** while maintaining 99.9% uptime and regulatory compliance.
- **Collaborated with global teams** to deliver end-to-end scalable cloud solutions using **GCP, Docker, Kubernetes**, and **CloudSQL**, enabling seamless **CI/CD** pipelines and reducing deployment time.
- **Mentored 5+ engineers** on system design, OOP principles, and cloud-native architecture, accelerating team onboarding and adoption.

Programmer Analyst Trainee **COGNIZANT** *Bengaluru, KA, IN* **08/2021 - 01/2022**

- Implemented Big Data solutions using Java, SQL, Hadoop, AWS for enterprise clients, improving data processing efficiency by 15%.
- Initiated a knowledge-sharing forum for new engineers, fostering collaboration and accelerating the adoption of new technologies.

Selected Projects

- **PyVisualizer (Open Source):** AST-powered code analysis engine transforming Python codebases into interactive architectural diagrams. **Impact:** Helping developers understand complex systems **10x faster**, adopted by 100+ developers globally.
- **PipelineIQ (Open Source):** An AI-powered platform providing predictive intelligence for BigQuery workflows. Features ML models that predict and optimize query performance with up to 84% accuracy.
- **Business Finance Management (HSBC):** Designed and implemented a scalable financial analytics system, significantly improving financial insights for **500K+ businesses**; architecture designed to be **cloud-agnostic (GCP & AWS adaptable)**.
- **MI Dashboard and Analysis (HSBC):** Created a comprehensive dashboard for business decision-making using BigQuery, Dataflow, and Looker, enabling real-time, data-driven financial insights with **0 error rate**.

Education & Continuous Learning

M.Tech in Applied Artificial Intelligence **Visvesvaraya National Institute of Technology, Nagpur** **01/2024 - 12/2025**

- *Professional & highly competitive master program from Top NIT of India*

B.Tech in Computer Science **SRM CEM** **08/2017 - 08/2021**

- *Won Best Project Award*

Awards & Achievements

- Top AI Innovator, Ford (2025) • Pat on the Back, HSBC (2024) • Rated Top Performer, HSBC (2023) • GCP Cloud Hero (10/2023)